

Sachin Kukkar

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EDUCATION

Swami Keshvanand Institute of Technology, M&G, Jaipur

Rajasthan, India

B.Tech in Computer Science and Engineering — CGPA 9.04/10, First Division with Distinction

2026

EXPERIENCE

Machine Learning Engineer Intern - ThoughtsWin Systems

September 2025 – June 2026

Computer Vision, Medical Imaging, PyTorch, Deep Learning

Jaipur, Rajasthan

- Built an end-to-end musculoskeletal (MSK) X-ray classification pipeline (Normal vs. Abnormal) in **PyTorch** on **58,000+** radiographs, benchmarking **EfficientNetV2**, **ConvNeXt**, and **Swin Transformer** architectures with AMP, AdamW, Cosine Annealing, Focal Loss, MixUp, CLAHE enhancement, and Test-Time Augmentation to reach **0.896 ROC-AUC**.
- Optimized **Faster R-CNN** and **Detectron2** pipelines for fracture localization, achieving a **98.23% F1-score** (97.57% precision, 98.90% recall) via ROI-based preprocessing and high-resolution training.
- Integrated **SAM2 / MedSAM** segmentation to improve anatomical localization and support explainable AI predictions, and built automated evaluation pipelines for ROC-AUC, F1-score, and confusion-matrix analysis.
- Contributed to **PleuraSight AI**, a deep learning system for pleural effusion detection on **NIH ChestX-ray14 (112K)** and **CheXpert (191K)** images, reaching **0.877 / 0.822 AUROC** in-domain / cross-institution with DenseNet-121 and Swin-B backbones.
- Collaborated with radiologists to translate clinical diagnostic workflows into ML pipeline design and trained models at scale on **NVIDIA H100** GPU infrastructure with reproducible, production-oriented codebases.

Intern - DRDO, Ministry of Defence, Govt. of India

May 2025 – July 2025

Certificate — Quantum Machine Learning

Delhi, India

- Selected as a **Quantum Computing Intern** at DRDO, working on simulation of key quantum algorithms.
- Simulated **Grover's**, **Deutsch-Jozsa**, and **QFT** using Python without Qiskit's Aer module.
- Built **NumPy-based simulators** to replicate quantum phenomena like superposition and entanglement.

Summer Trainee - Arootle Internet Pvt.Ltd.

July 2024 - August 2024

Certificate — Machine Learning, Python, NumPy, Pandas, Scikit-Learn

Jaipur, Rajasthan

- Gained hands-on experience with data cleaning, preprocessing, and exploratory data analysis (EDA) using Python libraries like Pandas, NumPy, and Matplotlib, improving the quality and usability of real-world datasets.
- Applied linear regression and classification techniques using Scikit-learn with feature scaling and data encoding, achieving 85% accuracy, 0.84 precision, 0.78 recall, and 0.81 F1-score in predicting fire occurrence on the Algerian forest fires dataset.

PROJECTS

EHSAAS | *TENS, Embedded Systems, Microcontrollers, Circuit Design, Sensors*

November 2022 – Present

- Founded and currently lead Project EHSAAS, developing a wearable device using **TENS** technology for menstrual pain relief, driving research, team collaboration, and prototype development to improve comfort and effectiveness.
- Secured an **INR 15 lakh** funding grant from MSME, Govt. of India, to advance R&D and bring innovation to menstrual health solutions.

IPC Section Suggestion System | *Machine Learning, Python, NumPy, Pandas, Scikit-Learn* July 2024 - August 2024

- Developed a machine learning system to suggest relevant **IPC Sections** from crime descriptions using text classification, TF-IDF feature extraction, and Logistic Regression / Naive Bayes models, achieving **87% accuracy** (0.86 precision, 0.84 recall, 0.85 F1-score). [Code](#).

TECHNICAL SKILLS

Languages and Frameworks: C/C++, Python, HTML, CSS, Bootstrap, Tailwind CSS

ML/DL & Computer Vision: PyTorch, Detectron2, Torchvision, OpenCV, Albumentations, Scikit-Learn

Developer Tools: VSCode, Postman, Git, GitHub, CUDA, Linux

Libraries: Pandas, NumPy, Matplotlib

Core Concepts: Deep Learning, Computer Vision, Object Detection, Model Evaluation & Optimization, Data Structures & Algorithms